Yushin Choi

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Research Interests

Versatile humanoid loco-manipulation through learning-based whole-body control, with a focus on generative models

Education

Hanyang University — Seoul, Republic of Korea
Master of Science in Intelligence and Information Engineering — *Mar. 2023 – Feb. 2025*

- Advisor: Prof. Wansoo Kim
- Thesis: Hierarchical Whole-Body Control for Enhanced Adaptability in Mobile Manipulators
- GPA: 4.33/4.5

Hanyang University ERICA — Ansan, Republic of Korea
Bachelor of Science in Robotics, summa cum laude (early graduation) — *Mar. 2017 – Feb. 2023*

- GPA: 4.12/4.5

Research Experience

Research Engineer | Robotics and Payload Control Systems Laboratory — Jan. 2025 – Present
LIG Defense & Aerospace — *Pangyo, Republic of Korea*

- **Guided Diffusion for Humanoid Whole-Body Control**
  - Implemented a cost-guided latent diffusion model for humanoid motion synthesis
  - Investigating conflicts between task guidance and dominant-mode bias when generating underrepresented motions
- **Hydraulic Full-Body Exoskeleton Control with Mission Payloads**
  - Developed gait-phase-dependent hydraulic force control for stable walking while carrying mission payloads
  - Developed FFT-based discrimination of human–exoskeleton interaction forces from external impacts for active balance stabilization

Graduate Researcher | Human-Robot Collaboration Research Laboratory — Mar. 2023 – Feb. 2025
Hanyang University (Advisor: Prof. Wansoo Kim) — *Seoul, Republic of Korea*

- **Whole-Body Control for Balancing a Quadruped Mobile Manipulator**
  - Designed and implemented a hierarchical whole-body controller with centroidal momentum regulation
  - Validated balance on a 15° slope under unknown object disturbances, where linear-only and no-momentum baselines failed
- **Learning-Based Whole-Body Manipulation with a Quadruped Mobile Manipulator**
  - Integrated an RL-based swing-up trajectory generator with whole-body tracking for a quadruped mobile manipulator
  - Identified feasibility gaps when applying low-dimensional RL-generated trajectories to whole-body robot control
- **Flexible Whole-Body Control Framework for Human-Robot Collaboration**
  - Developed a hierarchical whole-body controller with risk-driven continuous task-priority transitions
  - Validated tunable safety–comfort–efficiency trade-offs in blended coexistence and collaboration scenarios
- **Viewpoint-Aware VR Teleoperation with Force Feedback**
  - Developed viewpoint-aware pose and force transformations for intuitive VR teleoperation
  - Implemented virtual-robot inverse kinematics and haptic feedback through a physical leader manipulator

Undergraduate Research Project — Mar. 2022 – Feb. 2023
Hanyang University ERICA — *Ansan, Republic of Korea*

- **Multi-Robot Coordination for Office Assistance Services** (Advisor: Prof. Byung-Ju Yi)
  - Designed and implemented Firebase-based coordination of multiple Temi robots, including real-time communication and nearest-robot task allocation

Publications

Manuscript Under Review

[1] J. P. Jang*, Y. S. Choi*, D. G. Lee, and W. S. Kim, "Towards Flexible Human-Robot Collaborations: A Dynamic Control Framework Bridging Coexistence and Collaboration," manuscript under review. (*Equal contribution)

Peer-Reviewed Journal Articles

[1] D.-G. Lee, H. J. Kim, Y. S. Choi, S. Ha, J. No, S. Hwang, J.-K. Ryu, and W. Kim, "Posture Stabilization Control Framework for Lower Extremity Exoskeletons Operation in Protective-mission-oriented Tasks," *Journal of Institute of Control, Robotics and Systems*, vol. 32, no. 1, pp. 75–81, 2026.

[2] Y. S. Choi, J. P. Jang, S. M. Ha, S. W. Hwang, and W. S. Kim, "Whole-Body Control for Balancing an Articulated Legged-Arm Robot with the Integration of Momentum Management and a Passively Connected Unknown Object," *The Journal of Korea Robotics Society*, vol. 20, no. 1, pp. 112–119, 2025.

Conference Papers

[1] J. P. Jang, S. H. Hwang, Y. S. Choi, S. W. Hwang, and W. S. Kim, "Reinforcement Learning-Based Whole-Body Control for Dynamic Manipulation of Passively Connected Objects with a Legged-Arm Robot," *The Korea Robotics Society Annual Conference (KRoC)*, 2025.

[2] Y. S. Choi, S. H. Hwang, and W. S. Kim, "Whole-Body Control of an Articulated Legged-Arm Robot for Maintaining Balance with a Passively Connected Unknown Object," *The Korean Society for Precision Engineering (KSPE)*, 2024.

[3] J. P. Jang, S. M. Choi, Y. S. Choi, J. S. Lee, S. W. Hwang, and W. S. Kim, "Teleoperation Control Framework for Enhancing Operator Telepresence through Real-Time Virtual Environment Integration," *The Korea Robotics Society Annual Conference (KRoC)*, 2024.

[4] Y. R. Yoon, S. H. Lee, Y. S. Choi, S. R. Do, J. C. Park, J. Y. Yang, J. H. Woo, and B. J. Yi, "Development of Office Assistance Services Using Mobile Robots," *37th ICROS Annual Conference*, pp. 746–747, 2022.

Honors & Awards

Undergraduate Student Paper Award, 37th ICROS Annual Conference — 2022
University President's Award (2nd Place), Haedong Entrepreneurship Competition — 2021
Selected Team, Student Entrepreneurship 300 Program, Ministry of Education — 2021
Merit-Based Tuition Scholarship, Hanyang University ERICA — 2017, 2020 – 2022

Teaching Experience

Teaching Assistant — Sep. 2023 – Dec. 2024
Hanyang University ERICA — *Ansan, Republic of Korea*

- Developed laboratory materials and led hands-on sessions for a first-year robotics course
- Designed the final project and evaluated student hardware demonstrations

Entrepreneurship & Outreach

Early Core Team Member | DORO — Jul. 2020 – Feb. 2023
AI and Robotics Education Startup — *Ansan, Republic of Korea*

- Co-developed and iteratively refined DORO's early business model and robotics education strategy through pilot programs and service experimentation
- Developed and delivered robotics kits, curricula, workshops, and project-based programs for students across grade levels

Technical Skills

Programming: C/C++, Python, C#, MATLAB

Frameworks & Tools: PyTorch, Isaac Lab, ROS/ROS 2, WPF, Linux, Git

Simulation: Isaac Sim, RaiSim, MuJoCo, Gazebo, Unity

Robot Platforms: Franka Research 3, UR5e, Robotnik SUMMIT-XL STEEL, Temi, custom full-body exoskeleton

Hardware & Sensors: qb SoftHand Research, ATI AXIA-80 F/T sensor, Xsens MVN Awinda, OptiTrack